



LUMINAQ

The AI Pitch Decoder

10 Questions That Separate Real AI
Innovation from Expensive Demos

A Guide for Non-Technical Investors

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The Problem

*"You're in the pitch meeting. The founder says 'fine-tuned LLM with proprietary embeddings.'
You nod.
You understand nothing."*

If this sounds familiar, you're not alone. Most angel investors and family offices face the same challenge:

- Evaluating cutting-edge AI claims without a technical background
- No CTO on speed dial to validate founder claims
- Investment decisions based on vibes, decks, and demo energy
- The asymmetry: founders know more than you, and they know it

The result?

\$200B Wasted yearly on AI hype	85% Of AI startups fail to deliver	18mo Average runway before pivot
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This guide changes that.

How to Use This Guide

STEP 1

Review Before Any AI Pitch

Read through the 10 questions before or after any AI startup pitch meeting.

STEP 2

Ask Each Question

Use these questions during your evaluation. Note the quality and specificity of each response.

STEP 3

Score Each Area

Rate each response: Green (strong), Yellow (needs more info), or Red (warning sign).

STEP 4

Make Confident Decisions

Use the scorecard at the back to compile your assessment and make evidence-based decisions.

⚠ DECISION RULE

3 or more Red Flags = Walk away, or get expert review before proceeding. No exceptions.

01

QUESTION 01

“What specifically does your AI do that couldn’t be done with a simple rule-based system?”

WHY IT MATTERS

Many startups label basic automation as “AI” to justify higher valuations. This question forces founders to articulate genuine machine learning value—where the system learns, adapts, or handles complexity beyond what static rules can manage. If the answer is vague, the AI might just be a wrapper around simple if/then logic.

✓ GOOD ANSWER SOUNDS LIKE

“Our system processes unstructured radiology reports and identifies 47 anomaly patterns that evolve as new data comes in. A rule-based system would need manual updates for each new pattern—we’ve already adapted to 12 new patterns without human intervention.”

- ✓ Specific examples of learning/adaptation
- ✓ Quantified complexity that rules can’t handle
- ✓ Clear articulation of where ML adds value

⚠ RED FLAGS

- ✗ “Our AI is just better” with no specifics
- ✗ Can’t explain why ML is needed vs. simple automation
- ✗ The “AI” is essentially a decision tree or lookup table

FOLLOW-UP: *“Can you show me a specific case where the AI handled something a rule-based system would have missed?”*

02

QUESTION 02

“Where does your training data come from, and do you have the right to use it?”

WHY IT MATTERS

Data is the foundation of any AI system. If the training data is scraped without permission, biased, or too small, the AI's outputs will be unreliable. Legal exposure from improperly sourced data can destroy a company post-investment. This is the most common hidden liability in AI startups.

✓ GOOD ANSWER SOUNDS LIKE

“We have licensing agreements with three hospital networks covering 2.1 million anonymized records. Our data team includes a compliance officer who audits every new data source against GDPR and HIPAA requirements.”

- ✓ Named data sources with clear licensing
- ✓ Documented compliance processes
- ✓ Sufficient volume and diversity of training data

⚠ RED FLAGS

- ✗ “We scraped it from the internet”
- ✗ Vague about data sources or licensing
- ✗ Training set is suspiciously small for the claims being made

FOLLOW-UP: “Can I see your data licensing agreements and compliance documentation?”

03

QUESTION 03

“If OpenAI or Google released a similar feature tomorrow, what happens to your business?”

WHY IT MATTERS

This tests the startup’s true defensibility. If a major platform could replicate their core offering in a single release, the investment is essentially a bet that big tech won’t notice the opportunity—a bet you’ll usually lose. The best AI startups have moats that go deeper than the model itself.

✓ GOOD ANSWER SOUNDS LIKE

“Our moat isn’t the model—it’s 4 years of proprietary data from our 200+ enterprise clients, plus deep workflow integration that took 18 months to build per vertical. Google could build a similar model, but they’d need to replicate our data pipeline and industry relationships.”

- ✓ Acknowledges the threat honestly
- ✓ Points to moats beyond the model (data, distribution, integration)
- ✓ Has a clear answer, not a deflection

⚠ RED FLAGS

- ✗ Dismisses the threat entirely (“Google doesn’t care about our space”)
- ✗ Moat is solely “first mover advantage”
- ✗ Claims proprietary technology they can’t explain

FOLLOW-UP: *“Walk me through what a competitor would need to replicate—not just the model, but the full product.”*

04

QUESTION 04

“What are your model’s accuracy metrics, and how do you measure them?”

WHY IT MATTERS

Accuracy without context is meaningless. A model that’s 95% accurate sounds great until you learn the baseline (random guessing) is 90%. Understanding how metrics are measured—precision, recall, F1, against what benchmark—separates real AI from marketing AI.

✓ GOOD ANSWER SOUNDS LIKE

“On our test set of 50,000 labeled examples, we achieve 94.2% precision and 91.7% recall for fraud detection. The industry baseline using rule-based systems is 78% precision. We measure against a held-out test set that’s refreshed quarterly.”

- ✓ Specific numbers, not ranges
- ✓ Multiple metrics (not just “accuracy”)
- ✓ Comparison to a meaningful baseline

⚠ RED FLAGS

- ✗ Only reports “accuracy” with no context
- ✗ Can’t explain what the metrics mean practically
- ✗ Numbers change when you ask twice

FOLLOW-UP: *“What does a false positive cost your customer, and what does a false negative cost?”*

05

QUESTION 05

“When does your AI fail, and what happens when it does?”

WHY IT MATTERS

This is the honesty test. Every AI system fails—the question is whether the team knows where, how often, and what they’ve done about it. A founder who can’t describe failure modes either hasn’t tested properly or isn’t being straight with you. Both are disqualifying.

✓ GOOD ANSWER SOUNDS LIKE

“We see degraded performance with handwritten inputs—accuracy drops to about 72%. We’ve built a confidence scoring system that flags low-confidence outputs for human review. About 8% of cases go to manual review currently.”

- ✓ Specific failure scenarios described openly
- ✓ Confidence scoring or fallback systems in place
- ✓ Quantified failure rates

⚠ RED FLAGS

- ✗ “It doesn’t really fail”
- ✗ No monitoring or fallback system described
- ✗ Gets defensive when pressed on edge cases

FOLLOW-UP: *“Can you show me your error monitoring dashboard and how you handle edge cases in production?”*

06

QUESTION 06

“What is your technical moat—what would it take for a well-funded competitor to catch up?”

WHY IT MATTERS

This forces specificity about defensibility. Time, data, and domain expertise are real moats. “Our team is really smart” is not. You want to understand the specific barriers to replication, measured in months/years and dollars.

✓ GOOD ANSWER SOUNDS LIKE

“Replicating our system would require roughly 3 years of domain-specific data collection from 200+ manufacturing plants, regulatory approvals in 6 jurisdictions, and a team that understands both ML and industrial process engineering—a combination that’s extremely rare to hire for.”

- ✓ Quantified time/cost to replicate
- ✓ Multiple reinforcing moats (data + domain + distribution)
- ✓ Realistic assessment, not bravado

⚠ RED FLAGS

- ✗ Moat is “our proprietary algorithm” with no elaboration
- ✗ Claims years of lead time but product launched 6 months ago
- ✗ Can’t articulate barriers beyond team talent

FOLLOW-UP: *“If I gave a strong team \$10M and 18 months, where would they still fall short?”*

07

QUESTION 07

“How long did it take to go from prototype to working product, and what were the hardest engineering challenges?”

WHY IT MATTERS

This reveals technical honesty and maturity. If the team spent 2 years on a seemingly simple product, there are either deep technical challenges (good) or deep technical debt (bad). The nature of the challenges tells you whether the team has genuine ML expertise or is duct-taping APIs together.

 GOOD ANSWER SOUNDS LIKE

“Prototype took 3 months, but getting to production-grade accuracy took 14 months. The hardest parts were reducing inference latency below 200ms for real-time use and handling the long tail of edge cases in medical terminology.”

- ✓ Honest timeline with specific challenges
- ✓ Engineering challenges relate to real ML problems
- ✓ Clear distinction between prototype and production quality

 RED FLAGS

- ✗ “We built it in a weekend hackathon” for a complex product
- ✗ Challenges are all business-related, none technical
- ✗ Vague about engineering specifics

FOLLOW-UP: *“What’s the biggest technical compromise you’ve made, and what would you do differently with more resources?”*

08

QUESTION 08

“What are your unit economics for AI inference, and how do they scale?”

WHY IT MATTERS

AI infrastructure is expensive. GPU costs, API fees, and data processing can eat margins alive—especially as usage scales. If a startup can’t tell you what each API call or prediction costs, they haven’t thought about profitability at scale.

✓ GOOD ANSWER SOUNDS LIKE

“Each prediction costs us \$0.003 in compute. We charge customers \$0.15 per prediction, giving us a 98% gross margin on inference. At 10x current volume, our per-unit cost drops to \$0.001 due to batching optimizations.”

- ✓ Specific per-unit cost figures
- ✓ Clear relationship between cost and pricing
- ✓ Understanding of how costs change at scale

⚠ RED FLAGS

- ✗ “We’ll figure out margins later”
- ✗ Reliance on third-party API pricing they don’t control (e.g., OpenAI)
- ✗ No awareness of GPU/compute costs

FOLLOW-UP: *“What happens to your margins if your primary compute provider raises prices by 50%?”*

09

QUESTION 09

“Who on your team has actually built and deployed ML models in production before?”

WHY IT MATTERS

Research ML and production ML are vastly different disciplines. A team of PhDs who’ve only published papers may struggle with the messy reality of production systems. You want at least one person who’s shipped a real ML system that serves real users at scale.

✓ GOOD ANSWER SOUNDS LIKE

“Our CTO spent 5 years at Stripe building their fraud detection pipeline that processes 10M transactions daily. Our Head of ML previously led the recommendation system at Spotify for 3 years, serving 400M users.”

- ✓ Named individuals with specific production ML experience
- ✓ Experience at companies known for ML excellence
- ✓ Track record of shipping, not just researching

⚠ RED FLAGS

- ✗ All credentials are academic (PhDs, papers published)
- ✗ ML experience is limited to tutorials or Kaggle competitions
- ✗ “We’re hiring our ML lead” (meaning they don’t have one yet)

FOLLOW-UP: *“Can I speak directly with your technical lead about the architecture decisions they’ve made?”*

10

QUESTION 10

“Can you connect me with a customer who’s been using the product for at least 6 months?”

WHY IT MATTERS

This is the ultimate validation test. If a startup can’t produce a single reference customer with meaningful tenure, the product either doesn’t work well enough to retain users, or it’s too new to have been truly validated. Six months is the minimum to reveal real-world performance issues.

☑ GOOD ANSWER SOUNDS LIKE

“Absolutely. I’ll connect you with our VP of Operations at Acme Corp—they’ve been live for 11 months and recently expanded from one department to company-wide deployment. They can speak to both implementation challenges and ROI.”

- ✓ Immediate willingness to share references
- ✓ Customers with 6+ months of usage
- ✓ References who can speak to measurable outcomes

⚠ RED FLAGS

- ✗ Only offers very recent customers (< 3 months)
- ✗ Makes excuses about NDA restrictions
- ✗ “Our customers are too busy to talk”

FOLLOW-UP: *“I’d like to speak with a customer who almost churned or had a difficult implementation—can you share that reference too?”*

The AI Pitch Decoder Scorecard

Startup: _____

Date: _____

#	Question Area	Green	Yellow	Red
1	AI vs. Rules	☐	☐	☐
2	Training Data	☐	☐	☐
3	Core Dependency / Defensibility	☐	☐	☐
4	Accuracy Metrics	☐	☐	☐
5	Failure Cases	☐	☐	☐
6	Technical Moat	☐	☐	☐
7	Build Time / Maturity	☐	☐	☐
8	Unit Economics	☐	☐	☐
9	Team Depth	☐	☐	☐
10	Customer References	☐	☐	☐
TOTALS		—	—	—

NOTES:

DECISION FRAMEWORK

7+ Greens → Strong candidate — proceed with standard due diligence

4–6 Greens → Needs deeper technical due diligence before proceeding

3+ Reds → Walk away or get expert review — no exceptions

VERDICT: ☐ PASS ☐ INVESTIGATE ☐ NO

Quick Reference: 10 Questions

01

Why AI, not rules?

Tests: AI necessity & genuine ML value

Key red flag: *"AI is just better" with no specifics*

02

Training data source?

Tests: Data moat & legal compliance

Key red flag: *"We scraped it from the internet"*

03

What if Google ships this?

Tests: Defensibility beyond the model

Key red flag: *Dismisses platform risk entirely*

04

Accuracy metrics?

Tests: Technical rigor & honest measurement

Key red flag: *Only reports vague "accuracy"*

05

When does the AI fail?

Tests: Honesty & operational maturity

Key red flag: *"It doesn't really fail"*

06

What's the technical moat?

Tests: Barriers to replication

Key red flag: *Moat is "our secret algorithm"*

07

Prototype to product timeline?

Tests: Engineering depth & technical debt

Key red flag: *"Built it in a weekend" for complex product*

08

Unit economics at scale?

Tests: Path to profitability

Key red flag: *"We'll figure out margins later"*

09

Who's shipped production ML?

Tests: Team capability vs. credentials

Key red flag: *All PhDs, no production experience*

10

6-month customer reference?

Tests: Real-world validation

Key red flag: *Makes excuses about NDA restrictions*



PRO TIP: Question #5 (failure cases) is the single best honesty test. If a founder can't show you where the AI fails, they haven't tested it properly—or they're hiding something.

About the Author

Satish Rohit Singh

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With 15+ years of AI, Data Science, Platform Modernization, Enterprise IT and Data Architecture experience, Satish bridges the gap between complex AI technology and practical business decisions. As founder of LuminaQ, he provides technical due diligence services specifically designed for investors who need to evaluate AI startups without a technical co-founder on speed dial.

His approach is simple: strip away the jargon, test the claims, and give investors a clear verdict they can act on.

CREDENTIALS

- Corporate AI Mentor for Emirates Institute of Finance, Vinsys, Mindworx, Simplilearn, and The Knowledge Academy
- AI educator with a growing YouTube and Instagram following focused on practical AI expertise
- 15+ years in enterprise data architecture and machine learning infrastructure
- Published research in Springer Nature on XAI (Explainable AI).
- Based in Abu Dhabi, serving investors across the GCC and globally

CONNECT

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What's Next?

You now have a framework most angel investors never get. Here's how to use it:

PATH A: DO IT YOURSELF

Use this scorecard on your next 3 AI pitches. You'll be shocked how quickly you start spotting patterns you used to miss.

Print the scorecard. Bring it to meetings. Take notes. The questions do the heavy lifting.

PATH B: GET EXPERT BACKUP

Want a second opinion on a specific deal?

Book a 45-minute Pitch Audit. I'll review the deck with you, ask the questions founders hate, and give you a straight verdict.

[BOOK A PITCH AUDIT →](#)

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Questions? Reach out directly — I read every message.

THE AI PITCH DECODER



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